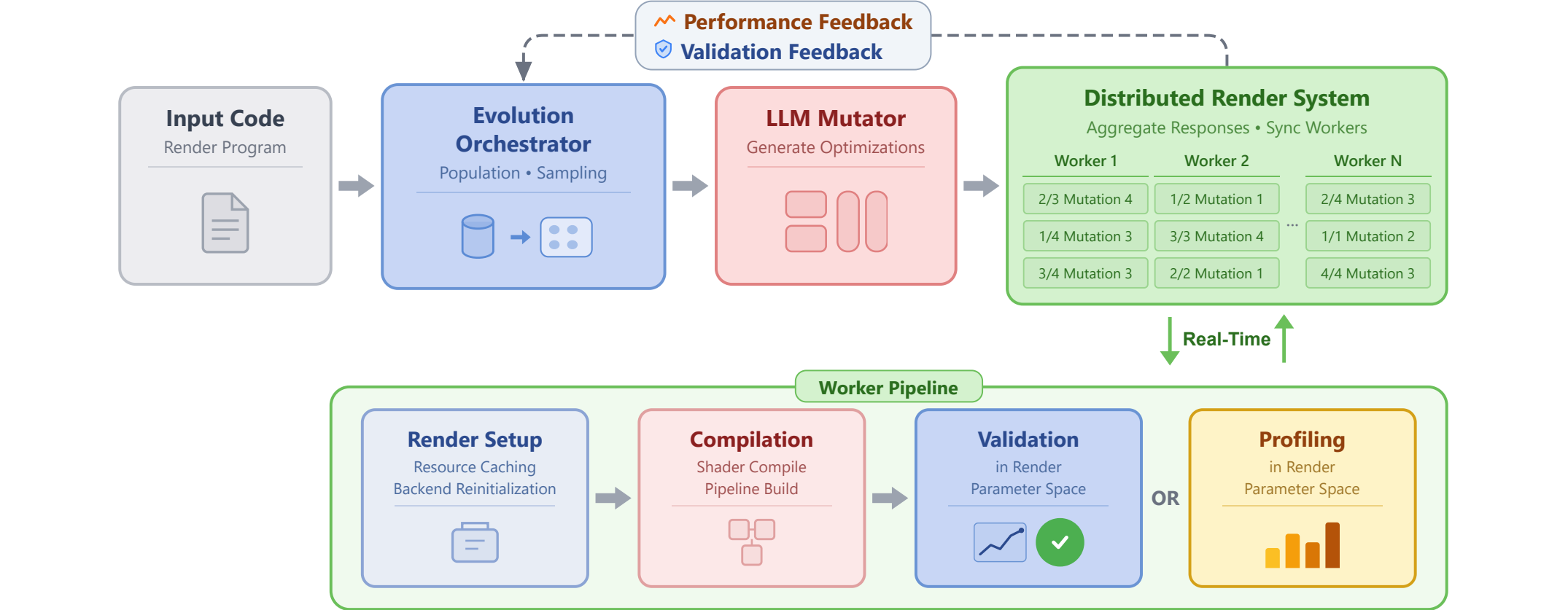




Mutations

Negligible FBM Removal

```
// SDF distance computation
- float fb = fbm(vec3(
-   10.1*x+offsetx, 10.1*y, 10.1*z));
- if( fb>0.0 )
-   di = di + 0.00000003*fb;
return sqrt(di);
+ // 3e-8 perturbation below float
+ // precision - removed entirely
+ return sqrt(di);
// ↑ Numerical reasoning: 36% boost
```

Chebyshev Polynomial

```
// Lighthouse stripe pattern
- cos(atan(pos.x,pos.z)*6.0)
+ float r_inv =
+   inversesqrt(dot(pos.xz,pos.xz));
+ float c = pos.z * r_inv;
+ float c2=c*c, c4=c2*c2, c6=c4*c2;
+ // T_6(x) = 32x^6-48x^4+18x^2-1
+ 32.*c6 - 48.*c4 + 18.*c2 - 1.;
// ↑ Math reasoning: atan+cos → polynomial
```

Beat-Conditional Guard

```
// Audio-reactive stage geometry
intersectStage(surface,pp,rot,scale);
- intersectStage(surface, pp,
-   rotateMat(angle.x, angle.y,
-     angle.z) * rot, scale);
+ float w = step(32.,beat)
+   * step(0., snare);
+ if (w > 0.5) {
+   mat3 r2 = rotateMat(angle.x,
+     angle.y,angle.z) * rot;
+   intersectStage(surface,pp,r2,scale);
+ }
// ↑ Domain reasoning: 40% boost
```